

EVANSTON, WY, USA

WMO: 725775

Lat: 41.273N

Lon: 111.031W

Elev: 7163

StdP: 11.27

Time Zone: -7.00 (NAM)

Period: 99-19

WBAN: 04111

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-0.7	2.9	-8.2	4.6	3.8	-3.9	5.8	8.4	28.6	27.9	26.4	28.0	8.1	230	0.706

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.9	86.5	55.4	84.0	54.5	81.6	54.0	58.6	77.0	57.3	76.6	56.0	75.5	13.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.4	79.4	61.8	50.9	72.4	61.3	48.4	65.7	61.6	28.5	76.6	27.5	76.1	26.6	75.1	63.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.6	24.6	21.3	DB	-8.1	90.1	5.0	2.0	-11.8	91.5	-14.7	92.7	-17.5	93.8	-21.2	95.2	
			WB	-8.8	60.7	4.9	1.6	-12.4	61.9	-15.2	62.8	-18.0	63.7	-21.6	64.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.0	22.1	23.6	31.6	39.8	48.8	59.2	68.5	65.9	57.0	44.2	32.3	21.9
	DBStd	18.27	8.84	8.76	8.77	8.37	8.53	7.94	4.44	4.86	7.87	9.07	10.47	9.25
	HDD50	4295	864	738	572	318	129	21	0	0	32	218	533	870
	HDD65	8259	1329	1158	1035	757	503	200	19	45	252	644	982	1335
	CDD50	1751	0	0	1	11	92	297	575	494	241	39	1	0
	CDD65	240	0	0	0	0	1	26	129	74	10	0	0	0
	CDH74	3331	0	0	0	0	24	457	1634	1007	205	4	0	0
	CDH80	848	0	0	0	0	2	98	514	214	20	0	0	0
	WSAvg	9.9	9.9	10.4	10.4	11.1	10.1	10.2	9.0	9.2	9.4	9.8	9.6	9.9
	PrecAvg	16.1	1.6	1.3	1.3	1.7	1.9	1.3	0.7	0.9	1.2	1.5	1.3	1.5
(15) Precipitation	PrecMax	23.0	3.0	2.5	2.9	3.9	5.0	4.2	1.9	2.2	3.5	4.2	2.4	3.7
	PrecMin	10.6	0.6	0.4	0.5	0.6	0.5	0.1	0.0	0.1	0.2	0.0	0.3	0.2
	PrecStd	3.1	0.8	0.5	0.6	0.7	1.1	1.0	0.5	0.6	0.7	0.8	0.6	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.2	48.2	61.2	69.9	77.0	86.9	90.0	87.7	82.2	72.6	61.3	46.1
		MCWB	35.6	35.5	41.6	46.2	51.4	55.5	57.0	55.5	52.9	47.6	42.0	35.2
	2%	DB	39.4	41.9	54.4	63.6	72.8	82.7	87.8	84.4	79.1	67.4	56.7	40.7
		MCWB	32.3	33.5	39.4	42.9	49.4	53.1	56.0	54.3	51.6	45.5	40.1	33.2
	5%	DB	36.5	38.7	49.2	59.7	69.7	79.4	85.5	82.3	76.2	63.6	51.3	37.0
		MCWB	31.1	32.4	37.0	41.6	48.3	52.3	55.4	53.9	50.2	44.0	37.6	31.2
	10%	DB	33.9	36.0	44.7	55.0	65.2	75.7	82.9	80.2	72.8	59.3	46.7	34.2
		MCWB	29.6	30.6	35.0	39.8	46.2	51.5	54.8	53.4	49.0	42.3	36.0	29.7
	0.4%	WB	36.7	37.3	42.8	47.2	53.4	57.5	60.5	59.4	57.0	50.3	44.0	37.0
		MCDB	42.0	45.0	57.9	68.0	73.5	80.8	77.3	76.5	73.2	63.7	57.2	43.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.1	35.0	40.0	44.2	51.0	55.4	59.2	57.9	54.4	47.6	41.0	34.3
		MCDB	37.8	40.0	53.0	60.4	68.8	77.1	77.2	75.9	70.9	62.3	52.5	38.9
	5%	WB	32.1	33.0	37.7	42.1	49.0	53.8	58.0	56.6	52.7	45.7	38.9	32.3
		MCDB	35.7	37.9	48.2	57.2	66.0	74.5	77.2	75.1	69.4	60.2	50.0	36.3
	10%	WB	30.0	31.1	35.3	40.5	47.2	52.4	56.8	55.3	50.8	43.6	36.7	30.2
		MCDB	33.2	35.5	43.9	53.8	63.2	72.5	76.6	73.9	67.8	57.3	45.8	33.8
(35) Mean Daily Temperature Range	5% DB	MDBR	14.4	14.4	16.5	20.6	22.2	25.6	26.9	26.6	25.1	21.7	18.0	14.0
		MCDBR	15.9	15.2	22.0	27.2	27.2	29.3	29.5	29.1	28.5	26.7	23.7	16.0
		MCWBR	11.5	10.2	11.9	13.1	12.3	12.3	10.9	10.9	11.8	12.6	12.6	10.8
	5% WB	MCDBR	14.0	13.7	20.8	25.1	24.8	26.7	25.4	25.7	24.6	24.2	21.6	14.7
		MCWBR	10.6	9.4	11.8	12.5	11.6	11.7	9.7	9.8	10.4	11.6	11.9	10.4
(40) Clear-Sky Solar Irradiance	taub		0.230	0.241	0.255	0.292	0.295	0.293	0.325	0.323	0.287	0.267	0.239	0.227
	taud		2.424	2.423	2.498	2.440	2.514	2.559	2.478	2.472	2.568	2.597	2.578	2.494
	Ebn at Noon		303	314	320	311	311	310	299	297	304	301	299	296
	Edh at Noon		24	29	30	35	33	32	34	33	28	24	21	21
(44) All-Sky Solar Radiation	RadAvg		677	954	1388	1752	2046	2449	2342	2035	1677	1161	750	554
	RadStd		57	84	99	101	178	174	117	96	116	82	78	56

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page